

# Python Console Expense Tracker

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## 1. COVER PAGE & OVERVIEW

A lightweight, command-line interface personal finance utility designed for instant offline tracking, fast calculations, and clean code architecture in Indian Rupees (INR).

## 2. INTRODUCTION & PROBLEM STATEMENT

**Introduction:** A minimalist offline CLI application to log, categorize, and manage daily expenditures without third-party libraries or external database overhead.

**Problem Statement:** Eliminates manual bookkeeping friction, tracking oversights, and the complexity of bloated finance software.

## 3. FUNCTIONAL & NON-FUNCTIONAL REQUIREMENTS

- **Functional:** Add expense records, view all entries, calculate cumulative totals, exit safely, and handle invalid inputs gracefully.
- **Non-functional:** Offline accessibility, zero dependencies, lightning-fast runtime execution speed.

## 4. SYSTEM ARCHITECTURE & DESIGN

**Architecture:** Event-driven procedural loop routing user choices through conditional branches.

**Data Schema:** In-memory list of dictionaries (expensesLIST holding expense key-value records).

## 5. IMPLEMENTATION DETAILS (CODE SNIPPET)

```
expensesLIST = [] while True: choice = int(input("1.Add 2.View 3.Total 4.Exit: ")) if choice == 1: date =
input("Date: ") cat = input("Category: ") desc = input("Desc: ") amt = float(input("Amount: ₹"))
expensesLIST.append({'date': date, 'category': cat, 'description': desc, 'amount': amt}) print("Expense added
successfully!") elif choice == 2: for idx, e in enumerate(expensesLIST, 1): print(f"{idx} - {e['date']} -
{e['category']} - ₹{e['amount']}") elif choice == 3: total = sum(e['amount'] for e in expensesLIST) print(f"Total
Expenses: ₹{total}") elif choice == 4: print("Exiting...") break else: print("Invalid Choice.")
```

## 6. TESTING & RESULTS

Verified menu routing, empty list behavior, and accurate currency summation through rigorous testing cases.

## 7. CHALLENGES & LEARNINGS

**Challenges:** Handling type conversion (`float`/`int`) and volatile runtime memory.

**Learnings:** Mastered Python loops, dictionaries, lists, and string formatting.

## 8. FUTURE ENHANCEMENTS

- Integrate JSON/CSV file storage for data persistence.
- Add robust exception handling for non-numeric inputs.
- Implement advanced category filtering.